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Avadi, Chennai

SCHOOL OF COMPUTING
DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
Academic year 2025-2026
(Winter Semester)

Course code: 10214AD701 **Course Name: Major Project**

TTS NO	Supervisor Name	VTU NO.	Student Name1	VTU NO.	Student Name 2
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Title of the Project: Enhanced Gastric Cancer Classification Using Optimized CNN Architecture.

Problem Statement

Gastric cancer is one of the leading causes of cancer-related deaths worldwide, primarily due to late diagnosis and the limitations of traditional manual analysis methods. Existing diagnostic systems rely heavily on expert interpretation of endoscopic and histopathological images, which can be time-consuming, subjective, and prone to errors. While recent deep learning approaches improve classification accuracy, most systems fail to provide quantitative information about the extent of cancer. Therefore, there is a need for an intelligent, automated, and efficient system that not only classifies gastric cancer accurately but also estimates the percentage of affected tissue to assist in better diagnosis and treatment planning.

Mapping of the Project to Complex Engineering Problem (CEP)

Level	Attribute	Characteristics	Targeted WKs (wherever applicable)	Justification (as applicable)
WP1	Depth of Knowledge	Requires strong knowledge in: • Deep Learning (CNN) • Image Processing • Medical Image	WK3, WK4, WK5, WK6, WK8	The project requires advanced understanding of CNN architectures, feature extraction, optimization techniques, and research-based medical imaging methods.

		Analysis Uses K3, K4, K5, K8		
WP2	Conflicting Requirements	Accuracy vs Computational Cost Model complexity vs efficiency Data quality vs availability	WK4, WK5, WK6, WK8	Balancing high accuracy with efficient computation and limited medical datasets is a major challenge.
WP3	Depth of analysis required	Have no obvious solution and require originality in analysis	WK3, WK4, WK5, WK8	There is no direct solution; requires: • Model design • Hyperparameter tuning • Performance optimization
WP4	Familiarity of issues	Involve infrequently encountered or novel problems	WK4, WK8	Medical image classification with percentage estimation is a **relatively novel problem**.
WP5	Extent of applicable codes	Address problems not encompassed by standards and codes of practice	WK6, WK8	No fixed standard exists for: • Cancer percentage estimation • CNN optimization in this domain
WP6	Stakeholder involvement and conflicting requirements	· Doctors · Patients · Researchers	WK5, WK6, WK7	Different stakeholders require accuracy, interpretability, and reliability.
WP7	Interdependence	Address high-level problems requiring a systems approach	WK3, WK5, WK6	System includes: • Data preprocessing • CNN model

				• Evaluation metrics

Mapping of the Project to Complex Engineering Activities (CEA)

EA No.	Attribute	Characteristics	Justification based on the Project
EA1	Range of Resources	Diverse resources and technologies	The project utilizes gastric cancer image datasets, deep learning frameworks such as TensorFlow/Keras, image processing tools like OpenCV, and GPU-based systems for efficient model training and evaluation.
EA2	Level of Interactions	Optimization of complex interactions	The system manages interactions between image preprocessing, feature extraction, CNN layers, and classification/regression outputs to ensure accurate prediction and efficient performance.
EA3	Innovation	Creative and research-based solutions	The project introduces an optimized CNN architecture with feature extraction and percentage estimation of cancer-affected regions, combining classification with quantitative analysis for improved diagnosis.
EA4	Consequences to Society and Environment	Significant societal impact	The system supports early detection of gastric cancer, improves diagnostic accuracy, and assists healthcare professionals in treatment planning, thereby contributing to better patient outcomes and reduced mortality rates.
EA5	Familiarity	Extends beyond prior experience	The project applies advanced deep learning techniques and medical image analysis, extending beyond conventional programming approaches and requiring understanding of emerging AI-based healthcare solutions.

Sustainable Development Goals (SDG) Mapping

SDG No.	SDG Title	Relevance to the Project
SDG 9	Industry, Innovation and Infrastructure	The use of advanced CNN models and AI-based medical image analysis promotes innovation in healthcare technology and supports the development of intelligent diagnostic systems.
SDG 11	Sustainable Cities and Communities	The system supports smart healthcare solutions by enabling efficient and reliable medical image analysis, contributing to improved healthcare services in modern communities.

Overall Justification: The project satisfies the characteristics of Complex Engineering Problems and Activities by requiring advanced knowledge in deep learning, medical image processing, and computer vision techniques. It addresses challenges such as accurate cancer detection, variability in medical images, and the need for quantitative analysis of cancer-affected regions. The proposed optimized CNN model applies research-based and innovative solutions to improve classification performance and provide percentage estimation of tumor regions.

Depth of Knowledge – Required Knowledge Levels (K3–K8)

The problem addressed in this project cannot be resolved without in-depth engineering knowledge spanning the following **Knowledge Profile levels**:

K3 – Engineering Fundamentals

· Understanding of image processing concepts such as filtering, normalization, and feature extraction

· Basics of machine learning and neural networks

· Mathematical foundations including matrices, probability, and statistics

K4 – Specialist Knowledge

- Advanced knowledge in cybersecurity techniques such as homomorphic encryption, intrusion detection systems (IDS), and zero-trust security models.
- Expertise in distributed databases, blockchain technology, and cloud-based system design.

K5 – Engineering Design

- Design of a multi-layered, microservices-based system architecture integrating security, data processing, and visualization layers.
- Development of role-based access control (RBAC) frameworks and secure data pipelines.

K6 – Engineering Practice

- Practical implementation using modern tools and frameworks such as Django, Node.js, React, PostgreSQL, MongoDB, and Hyperledger.
- Performance optimization, scalability testing, load balancing, caching strategies, and deployment in multi-node environments.

K7 – Engineering Comprehension

- Ability to analyze system behavior under cyberattacks, high user load, and large data volumes.
- Interpretation of system performance metrics, threat detection outputs, and visualization accuracy for decision-making.

K8 – Research Literature and Emerging Technologies

- Use of contemporary research in secure data analytics, privacy-preserving computation, blockchain auditing, and AI-driven threat detection.
- Evaluation and adaptation of recent IEEE research findings to design innovative and non-standard solutions.

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Signature

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(Supervisor Name)

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